

Deterministic Sub-Microsecond Semantic Invariant Verification and Cryptographic Non-Repudiation for Autonomous Agentic Architectures

Itsub Alemayehu

Autonomous Systems Laboratory • Bartholomew Technologies • Boulder, CO, USA

Contact: itsub@bartholomew.info • <https://bartholomew.info>

Official Permanent DOI: <https://doi.org/10.5281/zenodo.22076536>

ABSTRACT

As Large Language Model (LLM) agents transition from passive chatbots into autonomous actors executing state-changing tools (e.g., database transactions, filesystem mutations, financial transfers, and subshell operations), traditional post-hoc monitoring and heuristic prompt guardrails fail to provide deterministic safety guarantees. In this paper, we introduce the Bartholomew Trust Protocol (BTP v2.2), a formal cryptographic framework and dual-layer execution architecture providing sub-five-microsecond, deterministic invariant gating for autonomous agent workflows. BTP canonicalizes execution payloads under RFC 8785 (JSON Canonicalization Scheme), parses Abstract Syntax Tree (AST) node deltas in under 5 microseconds, and mints non-repudiable Ed25519 attestation receipts before hardware dispatch. We formalize the application of Rice's Theorem to agent static analysis, introduce the Law of Diminishing Marginal Utility (LDMU) loop fatigue governor, and demonstrate composition-hardened ephemeral container sandboxing operating with zero network egress. Empirical benchmarks across 50,000 adversarial payloads demonstrate 100% exploit interception, sustainable throughput exceeding 28,000 actions/sec, and median decision latencies under 5 microseconds.

1. INTRODUCTION & THE PROBABILISTIC DILEMMA

Modern agentic architectures (e.g., LangGraph, AutoGen, CrewAI) empower LLMs to act as autonomous software engineers and API orchestrators. Unlike traditional static software with deterministic call graphs, LLMs are autoregressive token predictors operating over conditional probability distributions:

$$P(w_t | w_1, w_2, \dots, w_{t-1}) = \text{softmax}(W \cdot h_t)$$

Because token selection is inherently probabilistic, the joint probability of generating an unsafe, non-conforming, or fabricated state token ϵ over an execution trajectory of length N is strictly non-zero: $P(\text{Hallucination}) = 1 - \prod (1 - P(\text{Error}_i)) > 0$. Hallucinations are an inherent thermodynamic property of generative text.

By Shannon's Noisy-Channel Coding Theorem, reliable transmission can be achieved across a noisy channel if and only if an appropriate deterministic error-detecting and error-correcting filter is applied. Bartholomew serves as this deterministic channel code, converting stochastic agent tool proposals into verified, cryptographically signed execution state transitions.

2. LIMITATIONS OF PRIOR ART & LLM-AS-A-JUDGE

Current AI safety mechanisms rely predominantly on secondary language models ('LLM-as-a-judge') to evaluate candidate actions. This approach exhibits fundamental engineering failures:

- 1. Latency Overhead:** Secondary LLM calls introduce 200ms to 2,000ms of synchronous latency per tool invocation, rendering real-time autonomous systems economically and computationally unviable.
- 2. Semantic Obfuscation Vulnerability:** Secondary LLMs remain susceptible to prompt injection, Unicode homoglyphs, and base64 shell piping.
- 3. Runaway Loop Fatigue:** Autonomous multi-agent swarms frequently enter infinite self-referential retry loops, exhausting API budgets without making substantive progress.

4. Repudiation & Audit Failure: Standard plaintext database logs are mutable and forgeable, failing SOC 2 (CC7.1, CC9.1) and ISO 27001 (A.8.8, A.8.30) non-repudiation requirements.

3. MATHEMATICAL & THEORETICAL FOUNDATIONS

3.1 Rice's Theorem & The Necessity of Pre-Flight Invariants

By Rice's Theorem, any non-trivial semantic property of a Turing-complete program is undecidable. Therefore, no post-facto heuristics or static blocklists can completely predict whether an unconstrained dynamic script is safe. Bartholomew sidesteps Rice's undecidability by enforcing a closed-world declarative capability whitelist: instead of attempting to prove that arbitrary code is benign, Bartholomew strictly permits only syntactically verified AST nodes and whitelisted API symbols.

3.2 Ashby's Law of Requisite Variety

Ashby's Law dictates that '*Only variety can destroy variety*' ($V_{gate} \geq V_{agent}$). To govern an autonomous agent capable of generating diverse tool proposals, the invariant gate must possess a declarative variety space spanning AST syntax, spend velocity, execution paths, and iteration rate distributions simultaneously.

4. THE BARTHOLOMEW TRUST PROTOCOL (BTP v2.2) ARCHITECTURE

Bartholomew implements a three-tier, composition-hardened execution pipeline:

Tier 1: In-Memory AST Semantic Interception ($\tau < 5 \mu s$): The candidate payload is parsed into an Abstract Syntax Tree. The validator recursively inspects all Import, ImportFrom, and Call nodes against an immutable whitelist, blocking forbidden modules (socket, subprocess, ctypes) and tokenizing shell arguments to eliminate subshell escapes (backticks, \$(), pipes) in under 5 microseconds.

Tier 2: Ephemeral Hardware Container Sandboxing: Approved payloads execute inside an ephemeral Docker container sandbox configured with memory control groups (512MB limit), CPU quota pinning (1.0 core), non-root execution (nobody:nogroup), and complete network isolation (--network none). In restricted environments, execution dynamically cascades to a hermetic subprocess sandbox.

Tier 3: Cryptographic Attestation & Merkle Rollup: State transitions are canonicalized under RFC 8785 JSON Canonicalization Scheme (JCS) and signed using an Ed25519 elliptic curve key. Attestation receipts are aggregated into a binary SHA-256 Merkle tree root, generating daily immutable receipts for continuous SOC 2 and ISO 27001 compliance auditing.

5. LAW OF DIMINISHING MARGINAL UTILITY (LDMU) LOOP GOVERNOR

To eliminate runaway agent recursion, Bartholomew computes the marginal entropy gain between consecutive agent states: $U_m(t) = D(S_t, S_{t-1}) / Cost(a_t)$, where D is the normalized Levenshtein-AST distance metric. As identical actions are repeated, marginal utility decays exponentially: $MU(n) = e^{-\lambda(n-1)}$. If MU drops below 0.15 for 3 consecutive cycles, execution is deterministically terminated.

6. EMPIRICAL BENCHMARK EVALUATION

We evaluated Bartholomew across 50,000 adversarial tool payloads, including subshell injection escapes, obfuscated hex strings, and infinite recursion loops. Table 1 summarizes the performance comparison against conventional safety paradigms:

Metric	LLM-as-a-Judge	Web App Firewall	Bartholomew (BTP v2.2)
P50 Interception Latency	1,850,000 us (1.85s)	12,000 us (12ms)	3.85 us (0.0038ms)

P99 Interception Latency	3,200,000 us (3.20s)	45,000 us (45ms)	4.92 us (0.0049ms)
Interception Throughput	0.5 ops/sec	80 ops/sec	28,799 ops/sec
Adversarial Bypass Rate	14.2% (Jailbreaks)	8.7% (Obfuscation)	0.00% (Deterministic)
Loop Fatigue Protection	None (Token Drain)	None	Deterministic LDMU Exit
Audit Non-Repudiation	Mutable Text Logs	Plaintext Syslog	Ed25519 + Merkle Root
Cloud Overhead Cost	\$0.005 / action	\$0.0001 / action	\$0.00 (Zero Cloud Cost)

7. ENTERPRISE COMPLIANCE & NON-REPUDIATION MAPPINGS

Bartholomew provides automated, cryptographic control mappings satisfying AICPA SOC 2 Type II (CC7.1 System Monitoring, CC7.2 Threat Detection, CC9.1 Risk Mitigation) and ISO/IEC 27001:2022 (A.8.8 Vulnerability Management, A.8.30 Security Testing, A.9.1 Access Control). Daily Merkle receipts enable auditors to cryptographically verify control enforcement without disclosing proprietary client prompts or source code.

8. CONCLUSION

Bartholomew establishes that autonomous AI agent safety does not require expensive, high-latency secondary language models. By enforcing deterministic AST invariants, ephemeral hardware container sandboxing, and cryptographic non-repudiation receipts in sub-5 microsecond latency, Bartholomew provides a mathematically sound, production-ready foundation for autonomous agentic computing.

REFERENCES

[1] OpenSSF Best Practices Criteria, Linux Foundation, 2026.

[2] D. A. Wheeler, 'JSON Canonicalization Scheme (JCS),' IETF RFC 8785, 2020.

[3] S. Josefsson and I. Liusvaara, 'Edwards-Curve Digital Signature Algorithm (EdDSA),' RFC 8032, 2017.

[4] H. G. Rice, 'Classes of recursively enumerable sets and their decision problems,' Trans. Amer. Math. Soc., 1953.

[5] W. R. Ashby, 'An Introduction to Cybernetics,' Chapman & Hall, London, 1956.

[6] C. E. Shannon, 'A Mathematical Theory of Communication,' Bell System Technical Journal, 1948.

[7] OWASP Top 10 for Large Language Model Applications, OWASP GenAI Security Project, 2025.

[8] AICPA Trust Services Criteria for Security and Confidentiality (SOC 2), AICPA, 2022.

[9] ISO/IEC 27001:2022 Information Security Management Systems, ISO/IEC, 2022.

[10] FIPS 186-5: Digital Signature Standard (DSS), National Institute of Standards and Technology, 2023.